

Limitless: Infinity

Repulsive-surface optimization for self-intersection-free meshes.

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Repulsive energies as self-intersection barriers.

Mesh interpolation, averaging, and packing routinely produce self-intersecting outputs. The tangent-point energy makes them impossible by construction.

Math lineage: knot energies (O'Hara, Strzelecki, von der Mosel) extended from curves to surfaces.

Cited applications: computational anatomy, soft-body robotics, biological membranes, digital manufacturing.

Discrete tangent-point energy.

Pairwise kernel; diverges at self-contact.

$$\hat{\Phi}(x) = \sum_{t_1 \neq t_2} a_1 a_2 \frac{|\langle n_1, c_1 - c_2 \rangle|^6}{\|c_1 - c_2\|^{12}}$$

Per-face centroids, normals, and areas.

Centroid c_t , unit normal n_t , area a_t , plus per-corner Jacobians. Rebuilt once per outer iteration; reused everywhere downstream.

Hierarchical $\mathcal{O}(n \log n)$ acceleration via BVH and block-cluster tree.

Admissible cluster pairs: $\max(\text{diam } U, \text{diam } V) \leq \theta \|c_U - c_V\|$.

Admissible $\rightarrow$ one rank-one cluster-aggregate term. Non-admissible $\rightarrow$ exact midpoint.

Brute-vs-hierarchical agree to machine precision at $\theta = 0$; expected $\mathcal{O}(\theta^2)$ decay otherwise.

Fractional Sobolev preconditioning at order $s = 5/3$.

$$\bar{A}^{-1} = (-\Delta)^{-1} L^{2-s} (-\Delta)^{-1}$$

Two cotan-Laplacian solves around one fractional matvec; applied via left-preconditioned GMRES.

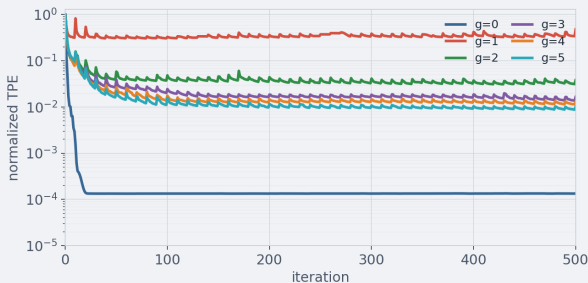
GMRES iteration count stays flat under mesh refinement.

Dynamic remeshing for triangle quality.

Per outer iteration: split edges $> \frac{4}{3}L_0$, collapse edges $< \frac{4}{5}L_0$, Delaunay flip, tangential smoothing toward the area-weighted circumcenter.

No-foldover guard rejects collapses that would flip a face normal.

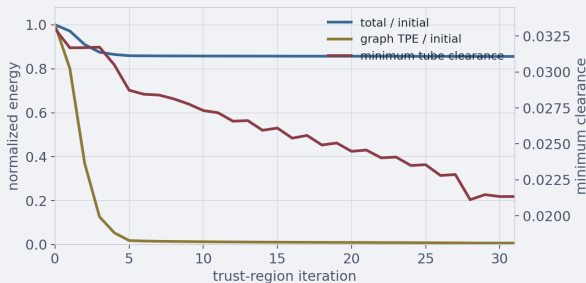
Canonical embeddings, genus 0 through 5.



Normalized TPE, 500 iterations, genus 0 through 5.

Genus 0 collapses to a sphere; higher genera plateau into symmetric local minima with handles intact.

Collision-free path optimization through a tube.



30-frame path, fixed endpoints.

Objective: shell elasticity +
self-repulsion + tube barrier.

Self-repulsion drops two orders of
magnitude in five iterations; no wall
contact.

Limitations.

CPU only

GPU port descoped; single-threaded.

Cloth eversion

Second Repulsive Shells hero demo not shipping; mid-surface init missing.

Future directions.

Karcher means

Multi-mesh averaging via the path solver.

Packing demo

Octopus-into-box (Repulsive Shells Fig. 21).

GPU port

Cluster-tree traversal parallelizes well.

Volumetric coupling

Embed thin repulsive shells in elastic FEM solids.

Summary.

Hierarchical tangent-point energy, fractional Sobolev preconditioned descent, dynamic remeshing, graph-manifold path solver. Two end-to-end demos.

Thanks. Happy to take questions.